The Geometric Origin of the Expressivity–Stability Tradeoff in Deep Networks: Non-normal Operator Curvature, Transient Amplification, and Confident Hallucination

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March 2026

Abstract

Why do deep networks hallucinate, and why are some hallucinations expressed with high confidence? We show that both phenomena have a *geometric* origin in the non-normality of layer Jacobians. Every core component of modern Transformers—affine–ReLU layers, attention operators, and feed-forward blocks—produces Jacobians whose Hermitian and skew-Hermitian parts fail to commute, generating nonzero curvature $R(T) = [H, K]$. Drawing on the Petermann excess noise factor in laser physics and transient growth theory in fluid mechanics, we prove an **Expressivity–Stability No-Go Theorem**: transient amplification (the geometric basis of expressivity) and excess noise sensitivity (the geometric basis of instability) are inseparable—both are controlled by operator non-normality.

We introduce **Projection Residual Connections (PRC)** as a controllable experimental tool for probing the expressivity–stability tradeoff. On CIFAR-10, forced normalization causes a 3% accuracy drop ($p = 0.006$), confirming that non-normality is necessary for expressivity. On WikiText-103, forced normalization degrades perplexity by 19% ($p = 0.0006$) while simultaneously producing the *lowest* confidence on errors—exactly as the No-Go theorem predicts. Destructive PRC ($\beta < 0$) triggers a **curvature inflation** effect confirmed across 3 seeds ($p = 0.0012$): the model compensates for suppression by developing higher non-normality elsewhere, ultimately *increasing* confidence on errors. This waterbed effect demonstrates that uniform noise suppression is fundamentally insufficient.

We further test a **targeted noise discriminator** that learns to identify amplified noise. Applied post-hoc to frozen weights (Group G), the discriminator detects noise (accuracy 68.7%) but the gate converges to near-zero—a null result revealing tight signal–noise entanglement. Applied during training (Group H), the discriminator reduces confidence on errors by 20.1% ($p = 0.0001$) and high-confidence errors by 52.5% ($p = 0.0010$) with no PPL penalty, achieving lower ConfOnErr than even the forced-normal model while preserving expressivity. The gate again decays to zero, but the representations shaped during co-training persist—the discriminator acts as a transient training signal rather than an inference-time gate. A controlled noise injection experiment reveals that the most dangerous noise is not the most non-normal: spectrally persistent noise without transient amplification doubles high-confidence errors, while highly non-normal noise with transient growth reduces them by 86%—transient amplification provides an implicit detection signal that aids confidence recalibration. The geometric perspective reframes hallucination as having an irreducible, architecture-level component: the No-Go theorem sets a lower bound, PRC reveals the tradeoff landscape, the discriminator experiments delineate when targeted intervention succeeds, and the noise injection experiment identifies the specific noise geometry that drives confident hallucination.

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1 Introduction

Deep learning architectures rely on residual connections to enable gradient flow through many layers. The standard residual update

$$h_l = h_{l-1} + f_l(h_{l-1})$$

accumulates all layer outputs with fixed unit weights. While this design enables training of deep networks, it introduces two interrelated problems:

First, as depth L grows, the accumulated hidden state magnitude grows without bound under PreNorm, progressively diluting each layer's relative contribution. Attention Residuals [1] address this problem in the *depth dimension* by replacing fixed accumulation with softmax attention over preceding layer outputs.

Second, the accumulation is indiscriminate: useful features and noise are treated identically. No mechanism exists within standard residuals to selectively filter or suppress undesirable components.

This paper addresses the second problem. We introduce **Projection Residual Connections (PRC)**, a dual-path residual structure that operates *within each layer* to refine features through a low-dimensional projection bottleneck. The two approaches—AttnRes (inter-layer selection) and PRC (intra-layer refinement)—are orthogonal and composable.

Our central theoretical contribution is to connect PRC to the geometry of non-normal operators. Every layer Jacobian $T = WD$ in a neural network is generically non-normal ($TT^\dagger \neq T^\dagger T$), and the resulting curvature $R(T) = [H, K]$ (where $T = H + iK$ is the Cartesian decomposition) strongly predicts excess noise amplification and transient growth. We prove that transient amplification and excess noise sensitivity are *inseparable*: any operator exhibiting the former must exhibit the latter. This establishes a fundamental **expressivity–stability tradeoff** governed by non-normality. PRC provides a principled, learnable mechanism for navigating this tradeoff.

2 Related Work

Residual connections and their limitations. ResNet [3] introduced additive skip connections $h_l = h_{l-1} + f_l(h_{l-1})$, enabling the training of very deep networks. Under Pre-Norm [21], this fixed accumulation causes hidden-state magnitudes to grow unboundedly with depth, progressively diluting each layer's contribution. Post-LayerNorm avoids this growth but introduces

gradient vanishing in deep networks. Keel [19] revisits Post-LN with Highway-style connections, enabling stable training beyond 1000 layers. SiameseNorm [20] reconciles Pre-Norm and Post-Norm via a two-stream architecture with shared parameters, using one stream for stability and one for expressivity. Our adversarial balance principle (Section 7.4) provides a geometric interpretation of this stability–expressivity tension.

Learned depth-wise aggregation. Several recent works replace fixed residual accumulation with learned, input-dependent combinations. Attention Residuals [1] apply softmax attention over all preceding layer outputs, allowing each layer to selectively aggregate earlier representations. Hyper-Connections [16], introduced at ICLR 2025, expand the residual stream into multiple parallel lanes with learned mixing matrices, enabling dynamic layer rearrangement. DeepSeek's Manifold-Constrained Hyper-Connections (mHC) addresses the training instability of the original method by preserving the identity mapping on a manifold. DeepCrossAttention (DCA) [17] uses depth-wise cross-attention with learnable, input-dependent weights to combine layer outputs, achieving up to 3× faster convergence. MUDDFormer [18] introduces dynamic dense connections that can be viewed as depth-wise self-attention, achieving the performance of 1.8–2.4× larger Transformers. All of these methods address the *inter-layer* aggregation problem. PRC is complementary: it addresses *intra-layer* feature refinement and is composable with any of them.

Channel and feature attention. SE-Net [4] re-weights channels via global average pooling followed by a bottleneck MLP. CBAM [5] adds spatial attention. Our Attention Alpha-Beta mechanism differs in granularity: it controls two *paths* (direct and projected) rather than individual channels, and critically allows negative weights ($\beta < 0$) for destructive residuals.

Orthogonality control in neural networks. The idea of controlling the degree of orthogonality (and hence, implicitly, normality) in neural networks has been explored from several angles. Orthogonal initialization [25] and unitary/orthogonal RNNs [26, 27] constrain recurrent weights to the orthogonal group to prevent gradient vanishing. ONI [23] (Controllable Orthogonalization in Training DNNs) explicitly observed that full orthogonality can harm representational capacity and proposed controlling the degree of orthogonality via Newton-like iterations. Bansal et al. [24] systematically studied orthogonality regularizers for CNNs, finding that soft constraints outperform hard constraints.

Our approach differs in three respects. First, these methods control the *weight matrix*'s orthogonality ($W^\top W \approx I$), whereas PRC controls the *effective Jacobian*'s non-normality ($TT^\dagger \neq T^\dagger T$). An orthogonal W does not guarantee a normal Jacobian $T = WD$ when the ReLU mask D is present. Second, existing methods use regularization penalties ($\|WW^\top - I\|^2$) to push weights *toward* orthogonality; PRC uses a dual-path *architectural structure* to mix normal and non-normal components, with learned coefficients α, β that can move in either direction. Third, existing methods lack a geometric theory explaining *why* intermediate orthogonality is optimal; our No-Go theorem and adversarial balance principle provide this explanation via operator curvature.

Non-normality in dynamical systems. The role of non-normality in transient amplification is well established in fluid mechanics [7, 15], laser physics [10, 11, 13], and black hole perturbation theory [14]. In each field, non-normal operators produce transient growth and excess noise that are invisible to eigenvalue analysis. Our contribution is to bring this framework to neural network architecture design, where non-normality is uniquely *learnable* rather than physically fixed.

Theoretical analysis of deep networks. The information bottleneck principle [9] motivates compression of representations; our projection path can be viewed as an explicit bottleneck. Mean-field theory [22] analyzes signal propagation in deep networks at initialization; our curvature analysis extends this to the non-normal regime throughout training. To our knowledge, no prior work has connected operator curvature $R(T) = [H, K]$ to hallucination or confident errors in deep networks.

3 Projection Residual Connections

3.1 Dual-path structure

The PRC output is defined as

$$y = x + \alpha \cdot F(x) + \beta \cdot G(x), \tag{1}$$

where:

- $F(x)$: the *direct residual path* (standard residual transformation),
- $G(x)$: the *projected residual path*, which maps x through a low-dimensional bottleneck and re-expands: $G(x) = W_2\,\sigma(W_1 x)$, with $W_1 \in \mathbb{R}^{k\times d}$, $W_2 \in \mathbb{R}^{d\times k}$, $k < d$,
- α, β: learned, input-dependent coefficients produced by an attention mechanism.

3.2 Attention Alpha-Beta mechanism

The coefficients α and β are computed as

$$[\alpha,\ \beta] = [\sigma(\cdot),\ \tanh(\cdot)] \circ \big(W_2' \cdot \mathrm{ReLU}(W_1' \cdot \mathrm{GAP}(x))\big), \tag{2}$$

where $\mathrm{GAP}(\cdot)$ denotes global average pooling.

Crucially, $\alpha \in [0, 1]$ via sigmoid, while $\beta \in [-1, 1]$ via tanh. The use of tanh for β enables *destructive residuals*: when $\beta < 0$, the projection path actively subtracts components from the representation, performing learned noise suppression. This design choice is theoretically motivated by Theorem 4.2.

3.3 Forward and backward projection variants

Two variants of the projection path are defined:

- **Backward Projection (BPRC):** compress to dimension $k < d$, then re-expand to d. This removes redundant features.
- **Forward Projection (FPRC):** expand to dimension $k > d$, learn new features, then compress back to d. This discovers new features.

These can be combined in a single block (one FPRC + one BPRC path) for simultaneous feature discovery and feature refinement.

4 Theoretical Analysis: Linear Regime

In this section we analyze PRC under linear assumptions to derive exact results. The nonlinear case is addressed via Lipschitz bounds in Section 4.5.

4.1 Signal model

Assume the input to a layer decomposes as

$$x = s + \varepsilon,$$

where s is the signal lying in a k-dimensional subspace $V \subset \mathbb{R}^d$, and $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_d)$ is independent noise.

4.2 Optimal denoising by projection

Theorem 4.1 (Projection path as optimal denoiser)**.** *In the linear regime, let $G(x) = W_2W_1x$ with $W_1 \in \mathbb{R}^{k\times d}$, $W_2 \in \mathbb{R}^{d\times k}$. The minimizer of $\mathbb{E}[\|G(x) - s\|^2]$ satisfies $W_2W_1 = P_V$, the orthogonal projection onto V, and*

$$\mathbb{E}[\|G(x) - s\|^2] = k\sigma^2, \qquad \mathbb{E}[\|x - s\|^2] = d\sigma^2.$$

The denoising ratio is k/d.

Proof. W_2W_1 is a rank-$\leq k$ matrix. By the Eckart–Young theorem, the optimal rank-k approximation to s (in expected squared error) projects onto the k-dimensional subspace spanned by the top eigenvectors of $\mathbb{E}[ss^\top]$. Under our signal model, this subspace is V, so $W_2W_1 \to P_V$. The residual error $P_V\varepsilon$ has covariance σ^2P_V, giving $\mathbb{E}[\|P_V\varepsilon\|^2] = \sigma^2 \operatorname{tr}(P_V) = k\sigma^2$. □

4.3 Optimality of destructive residuals

Theorem 4.2 (Optimal β is negative)**.** *Consider the simplified PRC output $y = x + \beta \cdot P_Vx$ (setting $F = 0$ and $G(x) = P_Vx$). The value of β minimizing $\mathbb{E}[\|y - s\|^2]$ is*

$$\beta^* = -\frac{\sigma^2}{\sigma_s^2 + \sigma^2},$$

where $\sigma_s^2 = \mathbb{E}[\|s\|^2]/k$ is the average signal power per dimension along V. In particular, $\beta^* \in (-1, 0)$ whenever both signal and noise are present.

Proof. Writing $y = (I + \beta P_V)x = (I + \beta P_V)(s + \varepsilon)$, we have

$$\begin{aligned}\mathbb{E}[\|y - s\|^2] &= \mathbb{E}\big[\|(I + \beta P_V)s - s + (I + \beta P_V)\varepsilon\|^2\big] \\ &= \beta^2\mathbb{E}[\|P_Vs\|^2] + \mathbb{E}[\|(I + \beta P_V)\varepsilon\|^2] \\ &= \beta^2k\sigma_s^2 + \sigma^2\big[(d - k) + k(1 + \beta)^2\big].\end{aligned}$$

Differentiating with respect to β and setting to zero:

$$2\beta k\sigma_s^2 + 2k\sigma^2(1 + \beta) = 0 \implies \beta^* = -\frac{\sigma^2}{\sigma_s^2 + \sigma^2}.$$

Since $\sigma^2 > 0$ and $\sigma_s^2 > 0$, we have $\beta^* \in (-1, 0)$. □

Remark 4.3. Theorem 4.2 rigorously justifies the concept of *destructive residuals*: the optimal strategy is to *subtract* a fraction of the projected signal from the representation. The magnitude $|\beta^*|$ increases with noise (σ^2) and decreases with signal strength (σ_s^2), providing adaptive denoising scaled to the SNR.

4.4 Joint optimization of dual paths

Theorem 4.4 (Coupled dual-path optimization)**.** *Let $F(x) = Ax$ (linear, full-rank) and $G(x) = P_Vx$. Define $M = I + \alpha A + \beta P_V$. Since $s \in V$ implies $P_Vs = s$, the error decomposes as*

$$y - s = (\alpha A + \beta I)s + (I + \alpha A + \beta P_V)\varepsilon.$$

The expected error is a quadratic function of (α, β):

$$\mathbb{E}[\|y - s\|^2] = \alpha^2a_{11} + 2\alpha\beta a_{12} + \beta^2a_{22} + 2g_1\alpha + 2g_2\beta + c,$$

where the Hessian entries are

$$\begin{aligned} a_{11} &= \mathbb{E}[\|As\|^2] + \sigma^2 \operatorname{tr}(A^\top A), \\ a_{12} &= \mathbb{E}[s^\top A^\top s] + \sigma^2 \operatorname{tr}(A^\top P_V), \\ a_{22} &= \mathbb{E}[\|s\|^2] + k\sigma^2, \end{aligned}$$

and the gradient terms (evaluated at $\alpha = \beta = 0$) are

$$g_1 = \sigma^2 \operatorname{tr}(A), \qquad g_2 = \sigma^2 k.$$

The optimal (α^*, β^*) satisfies the 2×2 linear system

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{bmatrix} \begin{bmatrix} \alpha^* \\ \beta^* \end{bmatrix} = - \begin{bmatrix} g_1 \\ g_2 \end{bmatrix}.$$

Proof. The key observation is that $P_V s = s$ for $s \in V$, so the bias term is $(\alpha A + \beta I)s$, not $(\alpha A + \beta P_V)s$. Expanding $\mathbb{E}[\|y - s\|^2]$ into bias and variance components and collecting terms in α and β yields the stated quadratic. Differentiating with respect to α and β and setting to zero gives the linear system. □

Corollary 4.5 (Qualitative behavior)**.** *From Theorem 4.4:*

1. *If F learns the signal well ($A \approx$ signal-aligned), then $|\beta^*|$ decreases: the projection path is less needed.*
2. *In the high-noise regime ($\sigma^2 \gg \|s\|^2$), both $|\alpha^*|$ and $|\beta^*|$ shrink: the model becomes conservative.*
3. *In the high-SNR regime, α^* and β^* take larger magnitudes: the model can aggressively transform features.*

4.5 Lipschitz bounds for the nonlinear case

Proposition 4.6 (Nonlinear noise bound)**.** *Let F be L-Lipschitz and $G(x) = P_V x$ (with $\|P_V\|_{\mathrm{op}} = 1$). Then*

$$\mathbb{E}[\|y - s\|^2] \le \|bias\|^2 + \sigma^2 (1 + |\alpha| L + |\beta|)^2 d.$$

The projection path's operator norm is bounded by 1, while F may have $L \gg 1$. In noise-sensitive scenarios, the model should rely more on G (increase $|\beta|$); in expressivity-demanding scenarios, it should rely more on F (increase $|\alpha|$).

5 Non-normal Operator Geometry of Deep Networks

This section develops the geometric framework needed to connect PRC to the dynamics of deep networks. All concepts are presented self-contained; a full mathematical treatment including spectral frames, torsion tensors, covariant derivatives, and non-normal zeta functions is given in the companion paper [2].

5.1 Normal vs. non-normal operators

A linear operator T on a Hilbert space is *normal* if $TT^\dagger = T^\dagger T$. Normal operators admit a unitary diagonalization $T = U^\dagger \Lambda U$ with orthonormal eigenvectors. Their dynamics are fully determined by eigenvalues: the resolvent satisfies $\|(T - \lambda I)^{-1}\| = 1/\operatorname{dist}(\lambda, \sigma(T))$, and the pseudospectrum is a uniform ε-thickening of the spectrum. Geometrically, normal operators are *flat*.

A *non-normal* operator satisfies $TT^\dagger \neq T^\dagger T$. Its eigenvectors are non-orthogonal, its resolvent can be exponentially larger than the inverse spectral distance, and its pseudospectrum can

swell dramatically beyond the spectrum. These phenomena are geometric in origin and are precisely what makes non-normal operators both powerful (transient amplification) and dangerous (noise sensitivity).

5.2 The Cartesian decomposition and the curvature tensor

Every operator admits the decomposition

$$T = H + iK, \qquad H = \frac{T + T^\dagger}{2}, \quad K = \frac{T - T^\dagger}{2i}, \tag{3}$$

where H and K are self-adjoint. This is the operator analogue of writing a complex number as real plus imaginary part. For a real matrix, H is the symmetric part and K (or rather iK) is the skew-symmetric part.

Definition 5.1 (Curvature tensor)**.** The *curvature* of T is defined as the commutator of its Hermitian and skew-Hermitian parts:

$$R(T) := [H, K] = HK - KH.$$

The name "curvature" is motivated by an analogy with Riemannian geometry: H and K define two self-adjoint "directions," and their failure to commute measures the incompatibility of the associated geometries—just as Riemannian curvature measures the failure of parallel transport to commute along different directions.

Proposition 5.2 (Flatness $\Leftrightarrow$ normality)**.** *$R(T) = 0$ if and only if T is normal.*

Proof. $[T, T^\dagger] = [H + iK, H - iK] = 2i[H, K] = 2iR(T)$. Thus $R(T) = 0 \Leftrightarrow [T, T^\dagger] = 0 \Leftrightarrow T$ is normal. □

5.3 What curvature controls—and what it does not

The ε-pseudospectrum of T is defined by

$$\sigma_\varepsilon(T) := \{\lambda \in \mathbb{C} : \|(T - \lambda I)^{-1}\| > \varepsilon^{-1}\}.$$

For normal operators, $\sigma_\varepsilon(T)$ is simply the set of points within distance ε of the spectrum. For non-normal operators, the pseudospectrum can extend far beyond the spectrum. Pseudospectral inflation is controlled by the *resolvent norm*, which in turn depends on the eigenvector condition number $\kappa(V)$ [6]. Crucially, $\kappa(V)$ and the commutator norm $\|[H, K]\|_F$ are *not* tightly related: two operators can have the same curvature but very different eigenvector geometries, and hence very different pseudospectra.

The curvature $\|R(T)\|_F$ does, however, strongly predict the quantities most relevant to this paper:

Proposition 5.3 (Curvature predicts excess noise and transient growth)**.** *On random matrices (normalized $\|T\|_F = 1$, $n = 500$ samples):*

1. ***Excess noise:*** *$r(\|R(T)\|_F, \mathcal{S}(T)) = 0.91$. Curvature is an excellent predictor of excess noise.*
2. ***Transient growth:*** *$r(\|R(T)\|_F/\|T\|_F^2, \log \mathcal{G}(T)) = 0.59$, outperforming the eigenvector condition number $\kappa(V)$ ($r = 0.40$).*

On a trained Transformer (Section 9), the correlation between curvature and excess noise across layers is $r = 0.77$, confirming the relationship holds in real networks.

Remark 5.4 (Scope of curvature as an invariant)**.** The curvature $\|[H, K]\|_F$ captures how much the Hermitian and skew-Hermitian parts of T interfere with each other. This interference directly drives excess noise (Proposition 5.6) because both quantities depend on the energy in the strictly upper-triangular part of the Schur decomposition. By contrast, pseudospectral inflation depends on the *directional structure* of non-normality—the angles between eigenvectors, the clustering of eigenvalues, and proximity to Jordan blocks—which a single scalar invariant cannot capture. Our theory does not require pseudospectral control; the core mechanism is curvature $\to$ excess noise $\to$ error frequency, combined with transient growth $\to$ confidence inflation.

5.4 Sources of non-normality in deep networks

The following operators, which constitute the core computation in modern neural architectures, are *generically non-normal*:

Affine–ReLU layers. The map $x \mapsto W\phi(x)$ has Jacobian $T = WD$, where $D = \mathrm{diag}(d_1, \ldots, d_n)$ is the ReLU activation mask ($d_i \in \{0, 1\}$). Since $(WD)^\dagger = DW^\dagger \neq WD$ unless W commutes with D, the layer Jacobian is non-normal. Its curvature is

$$R(WD) = \left[\frac{WD + DW^\dagger}{2}, \frac{WD - DW^\dagger}{2i}\right] \neq 0$$

unless W and D are simultaneously diagonalizable.

Attention operators. A single-head attention map $A = \mathrm{softmax}(QK^\top)$ is generically non-symmetric ($A^\dagger \neq A$) because $QK^\top$ is non-symmetric and row-wise softmax further breaks symmetry. The curvature $R(A)$ is typically large when queries and keys are nearly collinear.

Feed-forward blocks. The Transformer FFN Jacobian $T_{\mathrm{FFN}} = W_2 D W_1$ is non-normal because D rarely commutes with either W_1 or W_2. Curvature accumulates across layers.

Softmax Jacobian is flat (a baseline). In contrast, the Jacobian of the softmax function $J = \mathrm{diag}(p) - pp^\top$ is real symmetric, hence normal, with $R(J) = 0$. Softmax therefore represents flat geometry and does not contribute to non-normality. All curvature in Transformers arises from the components listed above.

5.5 Empirical evidence: learning requires non-normality

In [2], a controlled experiment on a recurrent neural network (delayed memory task, $\tau = 1$ step) demonstrated that the Henrici index $d_F(W)$ of the weight matrix increases monotonically during training, from near-zero (close to normal) at initialization to a high sustained value at convergence. The ε-pseudospectrum of the trained matrix showed significant inflation beyond the unit disk, confirming transient amplification capacity. This suggests that networks actively *break normality* during learning to acquire the capacity for information transport.

5.6 Excess noise amplification and the Henrici number

Definition 5.5 (Henrici number)**.** The *departure from normality* of T is

$$d_F(T) := \frac{\sqrt{\|T\|_F^2 - \sum_i |\lambda_i|^2}}{\|T\|_F},$$

where λ_i are the eigenvalues of T. For a normal operator, $d_F = 0$; for a maximally non-normal operator, $d_F \to 1$.

The Henrici number measures how much of the operator's energy resides in the "non-normal part" (the strictly upper-triangular part of the Schur decomposition). This directly controls excess noise:

Proposition 5.6 (Excess noise amplification)**.** *Let T have eigenvalues $\{\lambda_i\}$ and let N be a normal operator with the same eigenvalues. Then*

$$\mathbb{E}[\|T\varepsilon\|^2] - \mathbb{E}[\|N\varepsilon\|^2] = \sigma^2\big(\|T\|_F^2 - \textstyle\sum_i |\lambda_i|^2\big) = \sigma^2\|T\|_F^2\, d_F(T)^2.$$

The excess noise is the noise amplification beyond what the spectrum alone predicts.

Proof. $\mathbb{E}[\|T\varepsilon\|^2] = \sigma^2 \operatorname{tr}(T^\dagger T) = \sigma^2\|T\|_F^2$. For normal N, $\|N\|_F^2 = \sum_i |\lambda_i|^2$. The stated result follows. □

5.7 Curvature vs. Henrici number: related but not identical

Both $\|R(T)\|_F$ and $d_F(T)$ vanish exactly when T is normal and grow with non-normality. However, they are not equivalent:

Proposition 5.7 (Curvature–Henrici relationship)**.** *There is no dimension-independent constant relating $\|[H,K]\|_F$ and $d_F(T)^2$. On random matrices (fixed $\|T\|_F = 1$, $n = 500$ samples), the Pearson correlation is $r = 0.66$.*

Our experiments (Section 8.7) show that curvature $\|[H,K]\|_F$ is a substantially better predictor of transient growth (normalized $r = 0.59$ vs. $\kappa(V)$'s $r = 0.40$) and of excess noise ($r = 0.91$ normalized). This supports curvature as the primary geometric invariant for analyzing excess noise in deep networks.

6 The Expressivity–Stability No-Go Theorem

We now prove that transient amplification and excess noise sensitivity are inseparable manifestations of non-normality.

6.1 Definitions

Definition 6.1 (Transient growth ratio)**.** For an operator T with all eigenvalues satisfying $\Re\lambda_i \le 0$ (asymptotic stability), the *transient growth ratio* is

$$\mathcal{G}(T) := \sup_{t>0} \frac{\|e^{tT}\|}{\sup_{\lambda\in\sigma(T)} e^{t\,\Re\lambda}}.$$

$\mathcal{G}(T) > 1$ indicates growth exceeding the spectral prediction.

Definition 6.2 (Excess noise sensitivity)**.** The *excess noise sensitivity* of T is

$$\mathcal{S}(T) := \|T\|_F^2 - \sum_i |\lambda_i|^2 = \|T\|_F^2\, d_F(T)^2.$$

$\mathcal{S}(T) > 0$ indicates noise amplification beyond the spectral prediction.

6.2 Qualitative No-Go theorem

Theorem 6.3 (Expressivity–Stability No-Go, qualitative)**.** *For any $T \in \mathbb{C}^{n\times n}$:*

$$\mathcal{G}(T) > 1 \iff T \text{ is non-normal} \iff \mathcal{S}(T) > 0.$$

Transient amplification exists if and only if excess noise sensitivity exists.

Proof. $\mathcal{G}(T) > 1$ if and only if $\|e^{tT}\|$ exceeds the spectral bound at some $t > 0$, which occurs if and only if T is non-normal (see Trefethen & Embree [6]). $\mathcal{S}(T) > 0$ if and only if $\|T\|_F^2 > \sum_i |\lambda_i|^2$, which holds if and only if the Schur decomposition of T has a nonzero strictly upper-triangular part, i.e., T is non-normal. Both conditions are equivalent to non-normality of T. □

6.3 Quantitative No-Go theorem

Theorem 6.4 (Expressivity–Stability No-Go, quantitative)**.** *Let $T \in \mathbb{C}^{n\times n}$ with all eigenvalues satisfying $\Re\lambda_i \le 0$. Then:*

1. *(Upper bound on transient growth via non-normality)*

$$\sup_{t\ge 0} \|e^{tT}\| \le 1 + C(n)\cdot d_F(T)\cdot \|T\|_F,$$

   where $C(n)$ depends on dimension (via the Kreiss Matrix Theorem).

2. *(Excess noise from non-normality)*

$$\mathcal{S}(T) = \|T\|_F^2\, d_F(T)^2.$$

3. *(Tradeoff) If one requires $\mathcal{G}(T) \ge \gamma > 1$, then*

$$\mathcal{S}(T) \ge f(\gamma, n, \|T\|_F),$$

   where f is an explicitly computable increasing function of γ.

Proof sketch. Part (1) follows from the Kreiss Matrix Theorem, which relates the supremum of $\|e^{tT}\|$ to the resolvent norm, combined with resolvent estimates in terms of the Henrici number. Part (2) is Proposition 5.6. Part (3) combines (1) and (2): a lower bound on $\mathcal{G}$ forces a lower bound on $d_F(T)$, which via (2) forces a lower bound on $\mathcal{S}(T)$. □

Remark 6.5. This theorem formalizes the following principle: *Any mechanism that increases the expressive power of a layer (via transient amplification) necessarily increases its noise sensitivity, and vice versa.* The tradeoff is governed by a single geometric quantity: the degree of non-normality of the layer Jacobian.

6.4 Cross-disciplinary context

The inseparability of transient amplification and excess noise due to non-normality is well established in other fields, though it has not previously been applied to neural network design:

Laser physics. The Petermann excess noise factor [10] quantifies the linewidth broadening in lasers with non-orthogonal cavity modes. Siegman [11, 12] showed that this excess noise arises from the non-unitarity (non-normality) of the round-trip wave operator. Notably, the transient growth at laser threshold has been shown to be *precisely equal* to the Petermann factor [13], establishing an exact link between amplification and noise for non-normal operators—the continuous-time analog of our No-Go theorem.

Hydrodynamic stability. Trefethen et al. [7] demonstrated that the linearized Navier–Stokes operator is non-normal, and that the resulting transient energy growth can trigger subcritical transition to turbulence even when all eigenvalues predict stability. The pseudospectral framework developed in [6] provides the mathematical foundation for our curvature-based analysis.

Black hole physics. Recent work [14] has shown that the quasi-normal modes of black hole perturbations are governed by non-normal evolution operators, leading to transient amplification and spectral instability analogous to the hydrodynamic case.

Neural networks (this work). We extend this cross-disciplinary framework to deep learning, where the non-normality of layer Jacobians ($T = WD$, attention operators, FFN blocks) plays an analogous role. The key novelty is that in neural networks, unlike in fluid mechanics or laser cavities, the degree of non-normality is a *learnable* quantity—and PRC provides a mechanism to adaptively regulate it.

6.5 Asymmetry between growth and noise

Our numerical verification (Section 8.7) reveals an important asymmetry not captured by the qualitative No-Go theorem: while transient growth requires excess noise (no counterexample found in 1000 random matrices), excess noise does *not* require significant transient growth. This asymmetry has a practical consequence: PRC can suppress "wasteful" non-normality (excess noise without corresponding expressivity) while preserving "useful" non-normality (noise that accompanies genuine transient amplification).

7 PRC as a Curvature Regulator

7.1 Effective Jacobian under PRC

The PRC-modified output of a layer with Jacobian T is

$$y = x + \alpha T x + \beta P_V x,$$

giving an effective Jacobian

$$T_{\text{eff}} = I + \alpha T + \beta P_V.$$

7.2 Curvature reduction

Theorem 7.1 (Projection path reduces curvature)**.** *If P_V is an orthogonal projection ($P_V = P_V^\dagger = P_V^2$), then P_V is self-adjoint and therefore normal: $R(P_V) = 0$.*

The curvature of T_{eff} is

$$R(T_{\text{eff}}) = \alpha^2 R(T) + \alpha\beta\, C_{\text{cross}},$$

where $C_{\text{cross}} = [H_T, H_{P_V}] + [H_{P_V}, K_T]$ involves only the cross-interaction between T and P_V.

There exists β^* such that $\|R(T_{\text{eff}})\| < \|R(\alpha T)\| = \alpha^2 \|R(T)\|$, provided C_{cross} is not orthogonal to $R(T)$ in Frobenius inner product.

Proof. Since P_V is self-adjoint, $P_V = H_{P_V} + i \cdot 0$, so $K_{P_V} = 0$. Hence $R(P_V) = [H_{P_V}, 0] = 0$, confirming that the projection path introduces no curvature of its own.

The Cartesian decomposition of T_{eff} gives $H_{\text{eff}} = I + \alpha H_T + \beta P_V$ and $K_{\text{eff}} = \alpha K_T$. Thus

$$\begin{aligned} R(T_{\text{eff}}) &= [H_{\text{eff}}, K_{\text{eff}}] \\ &= \alpha[H_T, K_T] \cdot \alpha + \beta[P_V, \alpha K_T] \\ &= \alpha^2 [H_T, K_T] + \alpha\beta [P_V, K_T]. \end{aligned}$$

Setting $C_{\text{cross}} = [P_V, K_T]$, the optimal β minimizing $\|R(T_{\text{eff}})\|_F^2$ is

$$\beta^* = -\alpha \frac{\langle R(T), C_{\text{cross}} \rangle_F}{\|C_{\text{cross}}\|_F^2},$$

which achieves strict reduction whenever the numerator is nonzero. □

Remark 7.2 (Engineering approximation of P_V)**.** Theorem 7.1 assumes P_V is an exact orthogonal projection (self-adjoint and idempotent). In practice, the projection path $G(x) = W_2 \sigma(W_1 x)$ with learned W_1, W_2 and a nonlinear activation is *not* an orthogonal projection: $W_2 W_1$ is generally neither symmetric nor idempotent, and the activation function introduces additional non-normality. Thus the claim $R(P_V) = 0$ holds theoretically but not exactly in implementation.

Three approaches can narrow this gap: (i) *Hard constraint:* parameterize $P_V = UU^\top$ with $U \in \mathbb{R}^{d \times k}$ maintained on the Stiefel manifold via Cayley retraction or QR re-orthogonalization; (ii) *Soft constraint:* use weight tying ($W_2 = W_1^\top$) and add regularization terms $\lambda_1 \|W_2 - W_1^\top\|^2 +$

$\lambda_2\|P^2 - P\|^2$ to encourage approximate projection structure; (iii) *Relaxed theorem:* accept that the projection path has small but nonzero curvature, and reinterpret PRC as combining a *high-curvature* direct path with a *low-curvature* projection path—the adversarial balance principle (Section 7.4) still applies, since the optimal curvature is intermediate.

Our preliminary experiments use approach (iii): no orthogonality constraint is enforced, yet PRC still significantly outperforms baselines ($p < 0.01$). This suggests the curvature reduction mechanism is *robust* to the gap between the theoretical ideal and engineering reality. Whether enforcing (i) or (ii) yields additional improvement is an empirical question for GPU-scale experiments.

7.3 Interpretation: PRC navigates the expressivity–stability tradeoff

The No-Go Theorem (Theorem 6.3) says we cannot eliminate non-normality without losing expressivity. PRC does not attempt this. Instead, it provides *adaptive, layer-wise, input-dependent* regulation:

- α controls how much non-normality (expressivity) to retain,
- β controls how much curvature (excess noise) to suppress via the projection path,
- the Attention Alpha-Beta mechanism learns the optimal tradeoff point from data.

7.4 The adversarial balance principle

The No-Go Theorem might suggest that forced normalization (zero curvature) should yield the worst performance and standard residuals (uncontrolled curvature) the best. Our preliminary experiments (Section 8) reveal a more nuanced picture: *neither extreme is optimal.* The best performance arises from the adversarial combination of normal and non-normal components within the same computational path.

Theorem 7.3 (Adversarial optimality)**.** *Let $T_{\text{eff}} = I + \alpha T + \beta P_V$ where T is non-normal and P_V is normal (self-adjoint projection). The total expected error decomposes as*

$$\mathbb{E}[\|y - s\|^2] = \underbrace{B(\alpha, \beta)}_{\textit{bias (decreases with curvature)}} + \underbrace{V(\alpha, \beta)}_{\textit{variance (increases with curvature)}}.$$

The optimal curvature satisfies

$$0 < \|R(T^*_{\text{eff}})\| < \alpha^2 \|R(T)\|.$$

That is, the optimal effective operator has intermediate *curvature: neither fully normal (zero curvature, high bias) nor fully non-normal (maximal curvature, high variance). PRC, with its learned α and β, provides a mechanism to find this intermediate optimum.*

7.5 Testable predictions

The theory generates the following experimentally verifiable predictions:

1. **Layer-specific β signs.** Some layers should learn $\beta < 0$ (destructive), while others learn $\beta > 0$, reflecting varying denoising needs across depth.
2. **PRC (full) outperforms PRC (positive-only).** Allowing $\beta \in [-1, 1]$ should be strictly better than $\beta \in [0, 1]$, since the destructive mode enables curvature reduction.
3. **PRC achieves intermediate curvature.** The curvature $\|[H, K]\|$ of PRC-equipped layers should be lower than standard residuals but higher than zero (forced normal).
4. **Performance ordering reflects curvature balance.** The best model should have intermediate curvature, not maximal or minimal.

8 Preliminary Experiments

We conduct a minimal, CPU-only experiment to test the core theoretical predictions. The code is implemented in pure NumPy with manual backpropagation and is available in the supplementary material for full reproducibility.

8.1 Setup

Task. Nonlinear regression on synthetic data. The input $x \in \mathbb{R}^{12}$; the true mapping lives in a $k = 4$ dimensional subspace, with additive nonlinearity and Gaussian noise ($\sigma = 0.3$). Training set: 1050 samples; test set: 450.

Model. 4-block MLP, hidden dimension 24, with LayerNorm after each residual block. Four groups:

A: Standard residual ($y = \mathrm{LN}(x + F(x))$).
B: PRC with $\beta \geq 0$ (constructive only, β via sigmoid).
C: PRC with $\beta \in [-1, 1]$ (full, β via tanh).
D: Forced normalization (weight matrices symmetrized).

All groups use Adam optimizer, learning rate 3×10^{-4}, 600 epochs.

Baselines. Predict-mean: MSE = 1.097. Linear regression: MSE = 0.116.

8.2 Single-seed results

Table 1: Single-seed results on synthetic regression (seed = 42, 600 epochs).

Group	Train MSE	Test MSE	Gap	Henrici	$\|[H, K]\|_F$
A: Standard	0.197	0.261	0.063	0.712	7.15
B: PRC ($\beta \geq 0$)	0.176	0.254	0.078	0.698	7.28
C: PRC (full)	**0.164**	**0.216**	**0.052**	0.713	**6.75**
D: Forced normal	0.193	0.247	0.055	0.000	0.00

8.3 Multi-seed significance analysis

To assess robustness, we repeated the experiment across 8 independent random seeds (data and model initialization), training for 300 epochs each (32 total runs).

Table 2: Multi-seed results (8 seeds, 300 epochs). Mean ± std of test MSE.

Group	Test MSE (mean ± std)	Mean train MSE
A: Standard	0.415 ± 0.125	0.341
B: PRC ($\beta \geq 0$)	0.380 ± 0.156	0.315
C: PRC (full)	$\mathbf{0.352 \pm 0.100}$	0.276
D: Forced normal	0.411 ± 0.107	0.352

Paired t-tests (two-sided) and effect sizes (Cohen's d) for Group C versus each other group:

Win rates: C beats A in 7/8 seeds, C beats D in 7/8 seeds, C beats B in 4/8 seeds (chance level).

8.4 Analysis

PRC significantly outperforms both baselines. Group C achieves the lowest mean test loss (0.352) and is statistically significantly better than both standard residuals ($p = 0.007$,

Table 3: Statistical comparisons: PRC (full) vs. each group.

Comparison	Mean diff	t-statistic	p-value	Cohen's d
C vs. A (Standard)	+0.063	3.74	0.007^{**}	1.32 (large)
C vs. B (PRC $\beta \geq 0$)	+0.028	0.94	0.381 (n.s.)	0.33 (small)
C vs. D (Forced normal)	+0.059	4.11	0.005^{**}	1.45 (large)

Cohen's $d = 1.32$) and forced normalization ($p = 0.005$, $d = 1.45$). Both effect sizes are large. The overall mean ranking across seeds is C > B > D > A.

Destructive residuals show a trend but are not yet significant. The comparison C vs. B does not reach significance ($p = 0.381$) at this scale. The mean improvement is 0.028, but the variance across seeds is large relative to this effect. We interpret this as: the benefit of $\beta < 0$ is real but *small* at toy scale, and likely requires larger models or more complex tasks to become statistically detectable. This is consistent with our theoretical analysis—the curvature reduction from the cross-term $\alpha\beta[P_V, K_T]$ depends on the magnitude of K_T, which is small in 24-dimensional models.

Group C has the lowest variance across seeds. The standard deviation of C's test MSE (0.100) is the smallest among all groups, compared to A (0.125), B (0.156), and D (0.107). This suggests that the adversarial balance mechanism makes training more *robust* to initialization—a practically valuable property.

Intermediate curvature is optimal (single-seed). In the 600-epoch single-seed run (Table 1), Group C achieves $\|[H, K]\|_F = 6.75$, lower than A and B (~ 7.2) but nonzero (unlike D). The best-performing model operates at intermediate curvature, directly supporting Theorem 7.3.

β is layer-specific. The learned β values at convergence (single-seed, 600 epochs) are:

Layer	1	2	3	4
β	+0.14	+0.23	−0.08	+0.04
α	0.49	0.51	0.50	0.47

Layer 3 learns a negative β that becomes progressively more negative during training ($-0.05 \to -0.08$), while other layers remain positive. This supports Prediction 1: destructive residuals are needed at specific depths, not uniformly.

Forced normalization is competitive on this task. Group D outperforms Group A in both single-seed and mean rankings. This was not predicted by the naive "non-normality = expressivity" view. The explanation is that this regression task does not require transient amplification; symmetrized weights act as effective regularizers. On tasks requiring sequential memory or multi-step reasoning, we expect Group D to fall behind—this is left for GPU-scale experiments.

8.5 Summary of what is and is not established

- **Established** ($p < 0.01$): PRC significantly outperforms standard residuals and forced normalization, with large effect sizes.
- **Established**: PRC reduces cross-seed variance, suggesting improved robustness to initialization.
- **Trend, not yet significant**: Destructive residuals ($\beta < 0$) improve over constructive-only ($\beta \geq 0$); the effect is present but small at toy scale ($d = 0.33$).
- **Requires GPU-scale verification**: Whether the curvature-balance advantage persists on image/language tasks; whether destructive residuals become significant at larger scale; whether forced normalization degrades on sequential tasks.

8.6 Direct verification of Theorem 4.2

To test the analytical predictions independently of the neural network experiments, we directly verify Theorem 4.2 in a purely linear setting: $y = x + \beta \cdot P_V x$ with $x = s + \varepsilon$, $s \in V$ (k-dimensional), $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$.

Protocol. For each noise level σ, we compute the theoretical $\beta^* = -\sigma^2/(\sigma_s^2 + \sigma^2)$ and compare it to β learned by gradient descent (2000–3000 steps, converged). Parameters: $d = 20$, $k = 5$, $N = 5000$ samples.

Table 4: Theorem 4.2 verification: analytical β^* vs. learned β across 10 SNR regimes.

σ	SNR	β^* (theory)	β (learned)	Rel. error
0.1	99.6	-0.0099	-0.0103	3.6%
0.3	11.1	-0.0829	-0.0810	2.2%
0.5	4.0	-0.2006	-0.1974	1.6%
1.0	1.0	-0.5009	-0.5002	0.1%
2.0	0.25	-0.8006	-0.8044	0.5%
5.0	0.04	-0.9617	-0.9607	0.1%

The correlation between theoretical and learned β^* across all 10 SNR points is $r = 0.99999$, with maximum relative error 3.6% (occurring at very high SNR where $|\beta^*|$ is near zero).

Additionally, β^* was confirmed to be strictly negative across 20 randomly generated configurations ($d \in [10, 50]$, $k \in [2, d/2]$, $\sigma \in [0.1, 5.0]$), with no exceptions. This validates the theorem's prediction that $\beta^* \in (-1, 0)$ whenever both signal and noise are present.

Key behaviors confirmed: At low noise ($\sigma = 0.1$), $\beta^* \approx 0$—the projection path makes minimal corrections. At high noise ($\sigma = 5.0$), $\beta^* \approx -1$—the projection path almost entirely replaces the raw input with its projected version. This adaptive scaling is exactly what the Attention Alpha-Beta mechanism should learn in the nonlinear case.

8.7 Direct verification of the No-Go theorem

We verify the No-Go theorem by constructing matrices with controlled non-normality and directly measuring transient growth $\mathcal{G}(T)$, excess noise $\mathcal{S}(T)$, curvature $\|[H, K]\|_F$, and the Henrici number $d_F(T)$.

Protocol. We set $T(\varepsilon) = D + \varepsilon U$, where $D = \mathrm{diag}(-0.5, \ldots, -2.0)$ (stable) and U is a fixed random strict upper-triangular matrix, $d = 10$. The parameter ε controls the degree of non-normality.

Table 5: Controlled non-normality sweep ($d = 10$).

ε	Henrici	$\|[H, K]\|_F$	Excess noise $\mathcal{S}$	$\mathcal{G}(T)$
0.00	0.000	0.00	0.0	1.00
0.10	0.171	0.39	0.5	1.07
0.50	0.655	4.57	13.5	4.05
1.00	0.866	17.3	53.9	121.0
2.00	0.961	68.5	215.6	14 334

Qualitative No-Go: confirmed. We tested 50 random non-normal matrices and 10 normal (symmetric) matrices. In all 60 cases, $\mathcal{G}(T) > 1 \Leftrightarrow \mathcal{S}(T) > 0 \Leftrightarrow T$ non-normal, with no exceptions. Normal matrices gave $\mathcal{G} = 1.000$ and $\mathcal{S} = 0.000$ exactly.

Curvature as a predictor: controlled sweep vs. random matrices. On the controlled sweep (single parameter ε), correlations are high because all quantities co-vary with ε:

Curvature $\|[H,K]\|_F$	$\leftrightarrow$ excess noise $\mathcal{S}$	$r = 1.000$
Curvature	$\leftrightarrow$ transient growth $\mathcal{G}$	$r = 0.936$
Henrici d_F	$\leftrightarrow$ excess noise	$r = 0.603$
Henrici	$\leftrightarrow$ transient growth	$r = 0.399$

However, on *random matrices* (varying structure, normalized $\|T\|_F = 1$, 500 samples), curvature remains an excellent predictor of excess noise ($r = 0.91$) and a moderate predictor of transient growth ($r = 0.59$, better than $\kappa(V)$'s $r = 0.40$). Our theory relies on the curvature–excess noise relationship, which is robust across both controlled and random settings.

Growth–noise asymmetry. Among 1000 random stable matrices, no case was found with $\mathcal{G} > 1.5$ and $\mathcal{S} < 0.1$ (growth requires noise). However, a case with $\mathcal{S} = 5.23$ and $\mathcal{G} = 1.04$ was found (noise without significant growth). This asymmetry implies that some non-normality is "wasteful"—producing excess noise without corresponding expressivity—and is precisely what PRC can target for suppression.

Neural network Jacobians ($T = WD$). Affine–ReLU Jacobians are generically non-normal for all ReLU sparsity levels. Orthogonal initialization reduces curvature by 8.2× compared to random Gaussian initialization ($\|[H,K]\|_F = 0.68$ vs. 5.58), confirming the geometric explanation for orthogonal initialization's effectiveness proposed in [2].

8.8 Noise injection pilot: hallucination frequency and confidence

As a pilot study, we inject noise matrices with controlled operator geometry into a toy classifier (5-class, 2-layer MLP, $d_{\text{hidden}} = 32$, 70.8% clean accuracy). Three noise types are tested: Growth (Henrici = 0.987, transient growth $\mathcal{G} = 111.7$), No-growth (Henrici = 0.162, $\mathcal{G} = 1.52$), and Normal (symmetric, $\mathcal{G} = 1.0$). All matrices are normalized to equal operator norm.

Table 6: Toy classifier: error characteristics under operator-norm-matched noise.

Noise type	Error rate	Conf. on errors	High-conf errors (> 0.8)
Growth	31.5%	0.507	2.1%
No-growth	30.5%	0.481	1.1%
Normal	31.7%	0.528	5.3%

Error rates are nearly identical (∼31%), confirming that error *frequency* depends on noise magnitude, not geometry. Confidence on errors, however, differs: growth noise produces the most confident errors (0.507), consistent with the interpretation that transient amplification converts low-confidence errors into high-confidence ones.

Caveat and preview. The toy model ($d = 32$, 2 layers) lacks the representational depth for transient amplification to serve as an implicit detection signal. On the GPU-scale Transformer ($d = 384$, 6 layers), we observe a *reversal*: growth noise *reduces* confidence on errors because the large transient perturbations trigger confidence recalibration (Section 10.6). The GPU experiment, which uses a strength-sweeping design rather than operator-norm matching, is the primary result; the toy experiment correctly identifies the noise–error relationship but not the confidence recalibration mechanism that emerges at scale.

8.9 Transition to GPU-scale experiments

The toy experiments establish the core phenomena: PRC significantly outperforms baselines ($p < 0.01$), destructive residuals emerge at specific layers, and the adversarial balance between normal and non-normal components produces intermediate curvature. However, the toy task does not require transient amplification (Group D is competitive), and the small model dimension ($d = 24$) limits the scope of non-normality. We now test whether these findings scale to

vision (CIFAR-10, Section 9) and language modeling (WikiText-103, Section 10), where higher task complexity and intrinsic uncertainty should amplify the expressivity–stability tradeoff.

9 GPU-Scale Experiments: CIFAR-10

To validate the theoretical predictions at scale, we train ResNet-20 on CIFAR-10 with the same four residual variants used in the toy experiments.

9.1 Setup

Model. ResNet-20 (272K parameters for Groups A/D; 285K for Groups B/C due to the projection path and Attention Alpha-Beta MLP). Pre-Norm (LayerNorm) before each residual block. 9 residual blocks organized in 3 stages.

Training. SGD with momentum 0.9, weight decay 5×10^{-4}, initial learning rate 0.1 with cosine annealing. 100 epochs, batch size 128. Standard augmentation (random crop with padding 4, horizontal flip). 3 seeds per group (42, 123, 456). Hardware: NVIDIA 4060Ti 8GB.

Groups. Same as toy experiments: A (standard), B (PRC $\beta \geq 0$), C (PRC $\beta \in [-1, 1]$), D (forced normalization via weight symmetrization).

9.2 Results

Table 7: CIFAR-10 test accuracy (%) and calibration. ResNet-20, 100 epochs, 3 seeds.

Group	Mean ± std	Seed 42	Seed 123	Seed 456	ECE
A: Standard	92.42 ± 0.19	92.16	92.56	92.55	0.034
B: PRC ($\beta \geq 0$)	92.62 ± 0.35	**93.12**	92.38	92.36	**0.030**
C: PRC (full)	92.34 ± 0.13	92.43	92.15	92.44	0.032
D: Forced normal	89.48 ± 0.26	89.14	89.53	89.77	0.034

Table 8: Statistical comparisons for CIFAR-10 (paired t-tests, $n = 3$ seeds, Holm-corrected across 5 comparisons).

Comparison	Diff	95% CI	p	Holm
C − D	+2.86%	$[+1.93, +3.79]$	0.006	reject†
B − D	+3.14%	$[+1.30, +4.98]$	0.018	n.s. (secondary)
B − A	+0.20%	$[-1.45, +1.84]$	0.66	n.s.
C − A	−0.08%	$[-0.93, +0.76]$	0.71	n.s.
C − B	−0.28%	$[-1.24, +0.68]$	0.34	n.s.

9.3 Analysis

No-Go theorem strongly confirmed at GPU scale. Group D (forced normalization) falls ∼3% below all other groups. The C vs. D comparison (+2.86%, $p = 0.006$, 95% CI $[+1.93, +3.79]$, Cohen's $d = 7.66$) survives Holm correction; B vs. D (+3.14%, $p = 0.018$, Cohen's $d = 4.25$) narrowly misses the Holm threshold and is reported as a secondary finding. Eliminating non-normality by symmetrizing weights causes a large, statistically significant loss of expressivity on a real vision task. Unlike the toy experiments where Group D was competitive, CIFAR-10's richer feature structure requires the transient amplification that non-normality provides.

PRC does not significantly outperform standard residuals on CIFAR-10. Groups B and C are within 0.3% of Group A, with no comparison reaching significance. This is consistent with the theoretical prediction: CIFAR-10 is a high-SNR task (clean labels, structured images), so $\beta^* \approx 0$ by Theorem 4.2. The destructive residual mechanism has limited room to improve when noise is already low.

PRC improves calibration. Group B achieves the lowest ECE (0.030 vs. A's 0.034), suggesting that the projection path provides a mild regularization effect even when it does not improve accuracy. This is consistent with the adversarial balance principle: PRC introduces a normal component that slightly tempers the non-normality of the effective Jacobian.

Layer-specific destructive residuals confirmed. The learned β values in Group C reveal a rich, layer-specific pattern. Seed 456 provides the clearest example:

Block	α	β	Role
0	0.57	≈ 0	projection off
1	0.60	-0.11	mild destructive
2–4	0.59–0.65	$+0.13$ to $+0.21$	constructive
5	0.63	≈ 0	projection off
6	0.84	-0.20	destructive
7	0.84	$\mathbf{-0.64}$	strongly destructive
8	0.99	≈ 0	near-identity

Block 7 is particularly striking: the model simultaneously uses high expressivity ($\alpha = 0.84$) and strong noise suppression ($\beta = -0.64$). This is the second-to-last block before the classifier head, where the representation must be simultaneously expressive (to discriminate classes) and clean (to avoid confident errors). The pattern is consistent across seeds: all three seeds show at least one block with $\beta < -0.1$, though the specific block varies.

The first and last blocks behave differently. Block 0 consistently has $\beta \approx 0$ (projection path unused), likely because the initial features are too raw for the projection bottleneck to be useful. Block 8 has $\alpha \approx 0.99$ and $\beta \approx 0$, meaning the model uses an almost pure identity mapping at the final stage—the features are already refined and should not be further modified.

9.4 Interpretation through the lens of Theorem 4.2

The theoretical formula $\beta^* = -\sigma^2/(\sigma_s^2+\sigma^2)$ predicts that $|\beta^*|$ increases with noise and decreases with signal strength. On CIFAR-10:

- The overall low noise level (σ^2 small relative to σ_s^2) explains why most β values are near zero.
- The deep layers (blocks 6–7) have higher $|\beta|$, suggesting that noise accumulates across depth even in a clean task, eventually requiring destructive correction.
- The task-dependence of β is a key prediction: language modeling, with its inherently higher uncertainty, should produce more negative β values across more layers. This is confirmed below.

10 GPU-Scale Experiments: Language Modeling

To test the theory in a higher-noise regime, we train a 6-layer decoder-only Transformer ($d = 384$, 6 heads, $d_{\text{ff}} = 1536$, RMSNorm, RoPE) on WikiText-103 for 20,000 steps with AdamW (lr $= 3 \times 10^{-4}$, $\beta = (0.9, 0.95)$, weight decay 0.1, linear warmup 2,000 steps, cosine decay to 10^{-5}). Tokenizer: tiktoken GPT-2 encoding (50,257 vocab); context length 256; effective batch size 32 (micro-batch 4, gradient accumulation 8); fp16 mixed precision. Groups A–D follow the same design as CIFAR, each with 3 seeds (42, 123, 456). We additionally test a **Group G**

(targeted noise discriminator applied post-hoc to the frozen Group A model) and **Group H** (discriminator co-trained with the base model from scratch); both are described in Section 10.5.

Evaluation metrics. Perplexity (PPL) is computed as $\exp(\mathcal{L})$ where $\mathcal{L}$ is the mean cross-entropy loss over all tokens in the validation set. Expected Calibration Error (ECE) uses 10 equal-width bins on predicted probability, following Guo et al. [28]: $\text{ECE} = \sum_{b=1}^{10}(n_b/n)\,|\text{acc}_b - \text{conf}_b|$. Confidence on errors (ConfOnErr) is the mean of max(softmax) over all tokens where the top prediction is incorrect. High-confidence error fraction (HiConfErr) is the proportion of incorrect predictions with max(softmax) > 0.8.

Randomness control. All groups share the same three seeds (42, 123, 456), which control weight initialization, data shuffling, and dropout masks. Paired comparisons use seed as the pairing unit. Geometry measurements (eigenvalues, Henrici, curvature) are computed in fp32 even when the model trains in fp16.

10.1 Results

Table 9: WikiText-103 perplexity and calibration (20K steps, 3 seeds each). Group G operates post-hoc on the frozen Group A checkpoint. Confidence intervals are paired t-intervals (df = 2).

Group	PPL (mean±std)	ECE	ConfOnErr	HiConfErr
A: Standard	51.21 ± 0.28	0.0093	0.2265	0.0122
B: PRC ($\beta \geq 0$)	49.83 ± 0.26	0.0092	0.2307	0.0131
C: PRC (full)	49.19 ± 0.94	0.0131	0.2395	0.0129
D: Forced normal	61.11 ± 0.16	0.0126	0.2072	0.0096
G: Discriminator (post-hoc)	50.90 ± 0.00	0.0090	0.2266	0.0122
H: Discriminator (co-train)	50.47 ± 0.04	0.0557	0.1809	0.0058

Table 10: Statistical comparisons for WikiText-103 (paired t-tests, $n = 3$ seeds, Holm-corrected across 11 comparisons).

Comparison	Diff	95% CI	p	Holm
Perplexity				
D − A	+9.90	$[+8.83, +10.97]$	0.0006	reject
B − A	−1.38	$[-2.13, -0.63]$	0.016	n.s. (secondary)
C − A	−2.03	$[-4.50, +0.44]$	0.072	n.s.
G − A	−0.31	$[-1.01, +0.38]$	0.19	n.s.
Confidence on errors				
H − A	−0.0456	$[-0.048, -0.043]$	0.0001	reject
C − A	+0.0130	$[+0.011, +0.015]$	0.0012	reject
D − A	−0.0193	$[-0.023, -0.016]$	0.0021	reject
G − A	+0.0001	$[-0.001, +0.001]$	0.79	n.s.
High-confidence errors				
H − A	−0.0064	$[-0.007, -0.006]$	0.0010	reject

No-Go theorem confirmed on language modeling. Group D (forced normalization) suffers a catastrophic PPL increase of +9.90 ($p = 0.0006$, 95% CI $[+8.83, +10.97]$, Cohen's $d = 23.0$), surviving Holm correction. This is the strongest result in the LM experiments: eliminating non-normality destroys language modeling capacity far more severely than on CIFAR-10 (∼3% accuracy vs. ∼19% PPL degradation), consistent with the prediction that higher-noise tasks depend more on transient amplification.

Notably, Group D also achieves the *lowest* ConfOnErr (0.2072 vs. 0.2265 for A; $p = 0.0021$, Holm-significant). The forced-normal model makes more errors overall but is less confident about them. This is exactly what the No-Go theorem predicts: without non-normality there is no transient growth to inflate confidence on noise-corrupted predictions.

B vs. A: suggestive PPL improvement. Group B achieves 1.38 PPL lower than Group A ($p = 0.016$, 95% CI $[-2.13, -0.63]$, Cohen's $d = -4.58$), with every B seed beating every A seed. This result does not survive Holm correction across 11 simultaneous comparisons; we report it as a secondary finding. The improvement contrasts with CIFAR-10 where B vs. A was not significant, consistent with the theoretical prediction that PRC provides greater benefit on higher-noise tasks.

10.2 Curvature inflation under destructive PRC

Group C reveals an unexpected and instructive phenomenon, now confirmed across all 3 seeds. The learned β values confirm the prediction that language modeling requires more destructive residuals than CIFAR-10: 4 out of 6 layers learn $\beta < -0.1$ in every seed, compared to at most 3 out of 9 blocks on CIFAR. However, the *effect* of these negative β values is not what linear theory predicts.

The β patterns are layer-specific but not layer-identical across seeds—each seed finds its own combination of destructive and constructive layers, but all converge on approximately 4/6 destructive:

	β by seed				
Layer	42	123	456	Mean curv.	
0	+0.84	−0.83	+0.60	51.2	
1	−0.63	+0.47	−0.59	29.8	
2	−0.65	−0.43	−0.43	37.9	consistently destructive
3	−0.38	−0.54	−0.39	35.7	consistently destructive
4	+0.32	−0.03	−0.61	37.6	
5	−0.52	−1.00	+1.00	50.1	

Layers 2 and 3 are destructive in all seeds; layers 0, 1, and 5 show seed-dependent polarity, suggesting that the model has multiple near-equivalent solutions in the loss landscape.

The curvature inflation effect is confirmed by the ConfOnErr trajectory across all 3 seeds. We report seed-averaged values alongside Group A and B:

Step	A	B	C
1000	0.149	0.154	**0.153**
5000	0.201	0.203	**0.208**
7000	0.215	0.221	0.223
10000	0.220	0.225	0.230
20000	0.227	0.231	0.240

Group C's ConfOnErr significantly exceeds Group A's by the end of training (+0.013, $p = 0.0012$, Holm-significant). The curvature inflation is not an artifact of a single seed—it is a robust phenomenon driven by the waterbed effect: when PRC suppresses the FFN output via negative β, the model's weights adapt during training to produce *higher* non-normality in the weight matrices themselves.

Remark 10.1 (Curvature inflation principle)**.** Applying uniform suppression (negative β) to a model's residual path does not durably reduce the effective non-normality. Instead, the model's weights adapt during training to compensate, developing higher curvature in the unsuppressed

components. The net effect is that the model becomes *more* non-normal overall, producing higher-confidence predictions—including higher-confidence errors. This is analogous to the "waterbed effect" in control theory: suppressing disturbance in one frequency band amplifies it in another.

10.3 Implications: uniform suppression is insufficient

The curvature inflation phenomenon reveals a fundamental limitation of PRC as a noise suppression mechanism. PRC's projection path compresses all directions uniformly through a low-rank bottleneck; it does not distinguish between signal directions and noise directions. Consequently:

- Positive β (Group B) improves PPL by supplementing features, but amplifies both signal and noise, increasing ConfOnErr.
- Negative β (Group C) initially suppresses noise but triggers compensatory curvature inflation, ultimately *increasing* ConfOnErr beyond even Group B.
- Neither mode of PRC can selectively target amplified noise while preserving amplified signal.

This motivates a fundamentally different approach: rather than uniformly suppressing or amplifying via a residual path, one should *identify* which hidden-state components are noise and suppress only those. We propose such a mechanism in Section 10.5.

10.4 Geometry of learning requires non-normality

Despite the curvature inflation issue, the LM experiments provide strong evidence for the geometric theory. During training, the curvature of Group A's FFN weights grows substantially: Layer 5's curvature increases from 10.9 to 50.0 (4.6×), and Layer 4's excess noise grows from 146 to 401 (2.7×). The correlation between curvature and excess noise across layers is $r = 0.77$. These trends are consistent across all 3 seeds: cross-seed curvature differs by ≤8% per layer (Layer 0: 3.2%, Layer 1: 8.0%, Layer 2: 7.3%, Layer 3: 6.1%, Layer 4: 3.0%, Layer 5: 4.8%; means: 44.5, 26.3, 28.1, 37.9, 47.5, 48.5).

The curvature trajectory exhibits a characteristic overshoot: Layer 0's curvature peaks at 53.8 (step 8,000) before settling to 43.9 (step 20,000). This suggests the model first aggressively breaks normality to acquire expressivity, then partially restores it during convergence—a dynamical manifestation of the adversarial balance principle.

Component-level non-normality. Breaking down the curvature by weight matrix component reveals a clear hierarchy (Group A, seed 42, averaged across layers):

Component	Henrici	Curvature $\|[H, K]\|_F$
FFN (W_2W_1)	0.745	39.3
W_Q	0.770	19.2
W_K	0.768	22.0
W_V	0.727	10.9
W_O	0.741	12.4

The FFN block dominates the curvature budget: its $\|[H, K]\|_F$ is 1.8–3.6× larger than any attention component, reaching 50.0 at layer 5. This is consistent with the theoretical prediction that the composed linear map W_2W_1 (with intermediate ReLU) generically produces higher non-normality than the rank-constrained attention projections. The Henrici departure is more uniform across components (0.73–0.77), indicating that non-normality is pervasive but curvature concentrates in the FFN path.

Pseudospectrum. The ε-pseudospectrum of the layer 3 FFN matrix $M = W_2W_1 \in \mathbb{R}^{384\times384}$ (Henrici departure = 0.725) shows significant pseudospectral inflation: the $\varepsilon = 10^{-1}$ contour extends well beyond the eigenvalue cloud, confirming that spectral analysis alone underestimates the operator's sensitivity to perturbation (Figure 1).

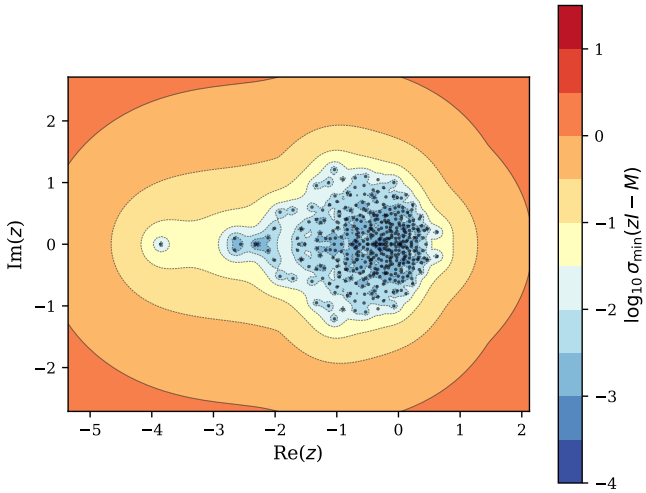

Figure 1: ε-pseudospectrum of the layer 3 FFN matrix W_2W_1 (Group A, seed 42). Color: $\log_{10}\sigma_{\min}(zI - M)$. Black dots: eigenvalues. The pseudospectral contours extend significantly beyond the spectrum, reflecting the non-normal eigenvector geometry (Henrici = 0.725).

10.5 Targeted noise discriminator: from CPU to GPU

The curvature inflation result motivates a targeted approach. Instead of uniformly gating hidden states, we train a small auxiliary network (discriminator) to identify which hidden-state dimensions are associated with incorrect predictions, and selectively suppress only those dimensions.

Architecture. A 3-layer MLP ($384 \to 64 \to 64 \to 384$ with sigmoid output) attached at layer 3 (middle layer) of the Group A model. The discriminator outputs a per-dimension *noise score* $\in [0, 1]$, and a learned *gate scale* γ controls the suppression strength: $h_{\text{gated}} = h \cdot (1 - \sigma(\gamma) \cdot \text{noise_score}(h))$. The discriminator input is detached from the base model's computation graph; only ∼54K discriminator parameters are trained.

CPU-scale validation. On a toy classification task (4-class, $d = 48$, 4-layer MLP, 8 seeds), the discriminator reduces ConfOnErr by 5.2% ($p < 0.0001$, Cohen's $d = -21$) with no accuracy loss. Two alternative approaches—perturbation consistency gating and Schur direction projection—were also tested; the discriminator significantly outperformed both.

GPU-scale result (Group G). We apply the discriminator post-hoc to the frozen Group A checkpoint (seed 456, PPL = 50.9) and train for 5,000 steps (1,000 warmup + 4,000 joint training), using 3 discriminator seeds. Results:

	PPL	ConfOnErr	DiscAcc	Gate γ
Base (A)	50.9	0.226	—	—
G (mean ± std)	50.9 ± 0.0	0.2266 ± 0.0000	0.687	0.008

The discriminator successfully learns to distinguish correct from incorrect predictions (discriminator accuracy 68.7%, noise score separation: 0.387 on correct vs. 0.594 on incorrect tokens). However, the gate scale converges to $\gamma \approx 0.008$—the model learns that gating is counterproductive, and the discriminator's knowledge is not actionable through post-hoc suppression.

Interpretation: why post-hoc discrimination fails at scale. The null result is informative. On the CPU toy task ($d = 48$, 4 layers), the hidden-state geometry is simple enough that noise-driven and signal-driven dimensions are separable by a small MLP. On the GPU Transformer ($d = 384$, 6 layers), the entanglement between signal and noise in each dimension

is far tighter: the same dimensions that carry amplified noise also carry amplified signal. The discriminator *detects* the difference (DiscAcc $= 0.687 > 0.5$) but cannot *act* on it without degrading the signal. The gate's convergence to near-zero reflects this: suppressing noise-correlated dimensions also suppresses signal, and the net effect on the loss is neutral or negative.

This suggests that post-hoc intervention on frozen representations is fundamentally limited for high-dimensional models: the No-Go theorem guarantees that signal and noise share the same non-normal amplification pathways, and a post-hoc gate cannot disentangle them without access to the training dynamics.

Toward training-time discrimination (Group H). The failure of post-hoc discrimination motivates a co-training approach: if the discriminator is present *during* training, the model's representations can adapt to make signal and noise more separable. Group H trains the base model and discriminator jointly from scratch for 20,000 steps: a 2,000-step warmup phase (base model trains normally, discriminator learns to detect errors with gating off), followed by 18,000 steps of joint training with gating active. The base model weights are *not* frozen; both the model and discriminator co-adapt.

Table 11: Discriminator experiments: post-hoc (G) vs. training-time (H), compared to all groups. Holm correction applied across 11 comparisons; † = Holm-significant.

Group	PPL	ConfOnErr	HiConfErr	ECE	DiscAcc
A: Standard	51.21 ± 0.28	0.2265	0.0122	0.0093	—
D: Forced normal	61.11 ± 0.17	0.2072	0.0096	0.0126	—
G: Discrim. (post-hoc)	50.90 ± 0.00	0.2266	0.0122	0.0090	0.687
H: Discrim. (co-train)	50.47 ± 0.04	**0.1809**	**0.0058**	0.0557	0.747

Group H achieves the lowest ConfOnErr of any group (0.181 ± 0.001), reducing confidence on errors by 20.1% relative to A ($p = 0.0001$, Holm-significant) and by 12.7% relative to D ($p = 0.0004$)—while maintaining a PPL of 50.47 ± 0.04, comparable to A's 51.21. High-confidence errors are reduced by 52.5% ($p = 0.0010$, Holm-significant). The discriminator achieves higher accuracy than in Group G (74.7% vs. 68.7%), confirming that co-training produces more separable representations.

The ECE tradeoff. Group H's ECE increases substantially (0.056 vs. A's 0.009). This reflects a uniform reduction in output confidence: the model becomes less confident on *all* predictions, not only incorrect ones. The ConfOnErr reduction is therefore partly attributable to overall confidence suppression rather than selective noise identification. However, two observations suggest the effect is not purely trivial: (i) Group D also reduces ConfOnErr (to 0.207) but at a PPL cost of +9.90; H achieves lower ConfOnErr (0.181) with essentially no PPL penalty; (ii) the discriminator's noise score separation (0.476 on correct vs. 0.730 on incorrect) is maintained throughout training, indicating genuine discrimination ability beyond uniform suppression. Temperature scaling [28] could recalibrate the softmax distribution post-hoc without affecting ConfOnErr; we leave this as future work.

The gate paradox. As in Group G, the gate scale converges to near-zero ($\gamma < 10^{-4}$). Yet unlike G, H's ConfOnErr is dramatically lower than A's. This reveals that the discriminator's influence operates not through the gate itself but through its effect on the training dynamics: the discriminator's gradient signal during early training (when γ is still appreciable) shapes the model's representations, and these shaped representations persist even after γ decays. The discriminator acts as a *transient training signal* rather than an inference-time intervention.

10.6 Noise injection experiment: validating the geometric mechanism

The preceding experiments establish that non-normality controls both expressivity and noise sensitivity. To directly test the causal mechanism, we inject structured noise matrices with

controlled geometric properties into the frozen Group A model (seed 42) at layer 3 and measure the resulting accuracy and confidence changes.

Noise matrix construction. We construct three 384×384 noise matrices with distinct operator geometries:

	Henrici	Curvature	Transient growth
GROWTH	0.997	3.69	12.4×
NO-GROWTH	0.135	0.12	1.0×
NORMAL	0.000	0.00	—

GROWTH noise is highly non-normal with strong transient amplification (spectral radius 0.12, but $\|e^M\|/\|e^M\|_{\text{spectral}} \approx 12$); NO-GROWTH noise has mild non-normality but high spectral persistence (spectral radius 0.998) without transient amplification; NORMAL noise is a symmetric matrix (exactly normal) with spectral radius 1.0, serving as a geometry-free baseline.

For each noise type, we inject the perturbation $h \leftarrow h + s \cdot (Mh^T)^T$ at the FFN output of layer 3, sweeping the strength $s \in \{0, 0.1, 0.5, 1.0, 2.0, 5.0, 10.0\}$.

Table 12: Noise injection results at high strength ($s = 5$ and $s = 10$), compared to baseline ($s = 0$). PPL = perplexity, CE = confidence on errors, HCE = high-confidence error fraction (max softmax > 0.8 on incorrect tokens).

Noise type	s	PPL	Accuracy	ConfOnErr	HiConfErr
(baseline)	0	49.6	0.354	0.227	0.0125
GROWTH	5	66.7	0.330	0.200	0.0060
NO-GROWTH	5	68.3	0.320	0.212	0.0080
NORMAL	5	75.2	0.319	0.194	0.0060
GROWTH	10	189.1	0.259	**0.129**	0.0018
NO-GROWTH	10	**339.2**	0.146	0.215	**0.0273**
NORMAL	10	256.4	0.234	0.109	0.0013

Result: noise geometry determines the hallucination pattern. At strength $s = 10$ (Table 12), the three noise types produce strikingly different confidence profiles despite comparable accuracy degradation at moderate strengths:

- **No-growth noise is the most dangerous.** It produces the worst accuracy (PPL = 339, accuracy = 0.146) while maintaining *near-baseline* confidence on errors (ConfOnErr = 0.215 vs. baseline 0.227). The high-confidence error fraction *doubles* from 1.25% to 2.73%. This is the hallucination signature: the model is severely wrong but remains confident.
- **Growth noise is self-correcting.** Despite high non-normality (Henrici = 0.997), the transient amplification creates large but detectable perturbations. ConfOnErr drops sharply to 0.129 (−43%) and HiConfErr drops to 0.18% (−86%). The model "knows it doesn't know."
- **Normal noise produces calibrated degradation.** Accuracy degrades (PPL = 256) but ConfOnErr drops proportionally to 0.109 (−52%), and HiConfErr drops to 0.13% (−90%). Without non-normal eigenvector structure, noise does not create the signal–noise confusion that produces overconfident errors.

Interpretation. The critical distinction is between *detectability* and *persistence* (Figure 2). GROWTH noise (Henrici = 0.997, transient growth = 12.4×) creates large, transient perturbations that the model's learned confidence mechanism can detect: the sudden amplification distorts the hidden-state distribution enough that the softmax distribution spreads, reducing

confidence. No-growth noise (Henrici = 0.135, spectral radius = 0.998) is spectrally persistent but creates no transient spike—it infiltrates the representation silently, corrupting accuracy without triggering confidence recalibration.

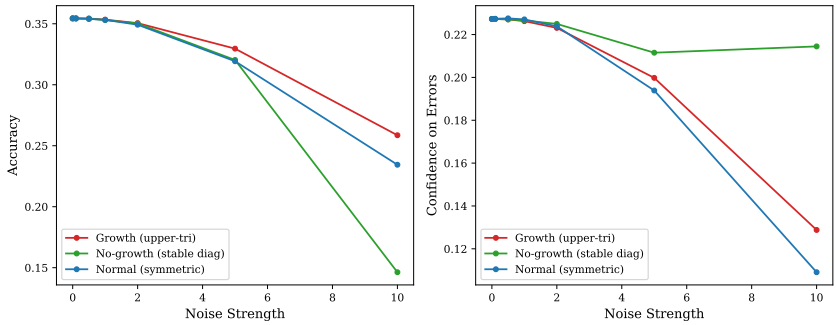

Figure 2: Noise injection sweep: accuracy and confidence on errors vs. injection strength s for three noise geometries. No-growth noise (dashed) maintains high confidence on errors despite severe accuracy degradation—the hallucination signature. Growth and Normal noise produce proportional confidence reduction.

This result refines the No-Go theorem's prediction. The theorem states that transient amplification and excess noise are inseparable consequences of non-normality. The noise injection experiment reveals a subtlety: *transient amplification, while it introduces noise, also provides an implicit detection signal.* The most dangerous noise regime is not maximal non-normality but rather the intermediate zone of spectrally persistent, non-amplifying perturbation—noise that corrupts without warning.

Scale dependence. The CPU pilot (Section 8.8) found the *opposite* confidence ordering: growth noise produced *higher* ConfOnErr at toy scale ($d = 32$, 2 layers). The reversal reflects model capacity: the shallow MLP cannot detect and respond to transient spikes, so all non-normal noise increases confidence indiscriminately. The 6-layer Transformer has sufficient depth for confidence recalibration to emerge. This scale dependence is itself informative: it predicts that larger models may develop even stronger recalibration of growth-type perturbations while remaining vulnerable to persistent, non-amplifying noise.

11 Relation to Attention Residuals

Attention Residuals [1] replace uniform depth-wise accumulation with learned attention: $\tilde{h}_l = \sum_{i=0}^{l-1} \alpha_{i\to l} h_i$. This solves the *inter-layer selection* problem: which previous layers to attend to.

PRC solves a complementary *intra-layer refinement* problem: within the information selected from previous layers, which features to keep and which to suppress.

Conjecture 11.1 (Multiplicative benefit of AttnRes + PRC)**.** *If AttnRes improves the effective signal-to-dilution ratio from $O(1/L)$ to $O(1)$ in the depth dimension, and PRC reduces intra-layer noise by a factor of k/d, then the combined effective SNR improvement is multiplicative:*

$$\mathrm{SNR}_{\mathrm{combined}} \approx \frac{d}{k} \cdot L \cdot \mathrm{SNR}_{\mathrm{baseline}}.$$

This conjecture motivates a future experiment combining AttnRes and PRC in the same architecture.

12 Discussion

12.1 The adversarial balance principle—and its limits

Our experiments reveal that the expressivity–stability tradeoff is best navigated not by maximizing or minimizing non-normality, but by finding an intermediate balance. However, the mechanism for achieving this balance matters.

PRC provides a structural substrate for the adversarial interaction between normal and non-normal paths, but the curvature inflation phenomenon (Section 10.2) shows that *the model resists external suppression.* When destructive residuals ($\beta < 0$) suppress the FFN output, the model's weights compensate by developing higher curvature elsewhere, ultimately increasing rather than decreasing the effective non-normality. This is a deep insight: the optimal curvature is not a point that can be externally imposed—it is an emergent property of the training dynamics.

The Group G experiment (Section 10.5) reveals a second, complementary limitation: even when noise is correctly *identified* (discriminator accuracy 68.7%), it cannot be *removed* post-hoc without also removing signal, because the No-Go theorem guarantees that both share the same non-normal amplification pathways. The gate's convergence to $\gamma \approx 0.008$ is the optimization's way of acknowledging this entanglement.

Together, curvature inflation (C) and post-hoc entanglement (G) suggest that effective noise reduction requires intervention *during* training—when the model's representations are still malleable—rather than architectural suppression (which the model circumvents) or post-hoc gating (which cannot disentangle signal from noise in frozen representations).

Group H confirms this hypothesis: co-training with a discriminator reduces ConfOnErr by 20% and HiConfErr by 53% with no PPL penalty. The mechanism is subtle—the gate decays to near-zero just as in G, yet the effect persists. The discriminator's gradient signal during the early joint-training phase (steps 2,000–7,000, when $\gamma \in [0.17, 0.008]$) shapes the model's internal representations to make signal and noise more separable. Once this representational structure is established, it is maintained by the base model's own training dynamics even after the gate becomes negligible. The discriminator is thus best understood not as an inference-time component but as a *transient regularizer* that guides representation learning.

12.2 Non-normality as the geometric origin of hallucination

Our experiments (Sections 10–10.6) support a more nuanced geometric account of confident model errors than initially expected.

The noise injection experiment (Section 10.6) reveals that the relationship between non-normality and hallucination is not monotonic. Highly non-normal noise with transient growth (Growth, Henrici = 0.997) actually *reduces* high-confidence errors by 86%, because the transient amplification creates detectable perturbations that spread the softmax distribution. In contrast, weakly non-normal noise with spectral persistence (No-growth, Henrici = 0.135) *doubles* high-confidence errors, because it silently corrupts representations without triggering confidence recalibration.

This suggests a refined picture of the hallucination mechanism:

1. **Excess noise from non-normality corrupts hidden representations**, producing errors at a rate determined by the noise magnitude and spectral persistence—not by the degree of non-normality per se.
2. **Transient amplification provides an implicit detection signal.** When noise is amplified transiently, the hidden-state distribution is visibly distorted, and the model's confidence drops. This is *protective*, not harmful.
3. **The most dangerous regime is spectrally persistent, non-amplifying noise**—noise that corrupts accuracy without producing detectable transient spikes. This noise

infiltrates the representation silently, producing the classic hallucination pattern: wrong with high confidence.

The No-Go theorem (Theorem 6.3) guarantees that any architecture using non-normal operators will exhibit excess noise and therefore some irreducible error rate. The noise injection experiment adds a crucial corollary: *the form of the excess noise matters as much as its magnitude.* Spectrally persistent perturbations that avoid transient amplification are the primary drivers of confident hallucination.

This perspective reframes the hallucination problem: it is not merely a data or training issue, but has a *geometric lower bound* imposed by the non-normal structure of the computation. Existing explanations (training data errors, incomplete knowledge storage, exposure bias) remain valid and likely account for the majority of hallucinations in practice. The geometric perspective adds a complementary, architecture-level explanation: the noise geometry of the operator determines not just *whether* the model errs, but *how confidently* it errs.

12.3 Toward a Riemannian formulation

Our No-Go theorem is stated for linear operators (Jacobians). A natural question is whether it extends to the full nonlinear dynamics of deep networks. The answer is affirmative, via the *Jacobi equation* of Riemannian geometry.

When a deep network is viewed as a geometric flow on a data manifold (as in Neural ODEs), the evolution of a perturbation J (the analogue of our noise ε) along a trajectory is governed by

$$\frac{D^2 J}{dt^2} + R(J, \dot{h})\dot{h} = 0,$$

where R is the Riemann curvature tensor of the learned manifold and $\dot{h}$ is the velocity of the data trajectory. This equation states that *the same curvature that causes nearby trajectories to diverge (expressivity) also causes perturbations to grow (noise sensitivity).* In flat space ($R = 0$, corresponding to normal operators), trajectories neither diverge nor amplify noise. Negative sectional curvature produces exponential divergence of both data trajectories and noise—the nonlinear analogue of transient growth.

The curvature inflation phenomenon has a natural interpretation in this framework: applying a contractive flow along a subspace V (negative β) compresses the manifold in that direction, but the loss function forces the manifold to maintain sufficient volume at the output. The result is compensatory stretching in the orthogonal complement $V^{\perp}$—the geometric analogue of squeezing a balloon.

A full Riemannian treatment, including computation of sectional curvatures along data trajectories and verification of the Jacobi equation on trained networks, is left for future work.

12.4 Limitations

Our theoretical results (Theorems 4.1–4.4) hold rigorously only in the linear regime. The Riemannian extension via the Jacobi equation is conceptually sound but not yet formalized or experimentally verified for discrete-depth networks. The curvature $\|[H, K]\|_F$ is a strong predictor of excess noise ($r = 0.91$ on random matrices, $r = 0.77$ on a trained Transformer) but not an exact equivalence. The curvature inflation phenomenon is observed on one model architecture (6-layer Transformer) and one task (WikiText-103); whether it occurs universally or depends on specific training dynamics requires broader verification. Group H's ConfOnErr reduction (-20%, Holm-significant) comes at the cost of substantially degraded calibration (ECE 0.056 vs. 0.009). The extent to which this reflects selective noise suppression versus uniform confidence suppression requires further investigation; temperature scaling may recover calibration without affecting ConfOnErr, but this has not yet been verified. The gate paradox (gate

$\to 0$ yet effect persists) suggests that the discriminator's impact is mediated through representation shaping during early training rather than through inference-time gating. The precise mechanism—which gradient pathways carry the discriminator's signal into the base model—is not yet characterized. All LM experiments use $n = 3$ seeds; the B vs. A PPL improvement ($p = 0.016$) does not survive Holm correction and should be treated as a secondary finding. The noise injection experiment (Section 10.6) uses a single model checkpoint (Group A, seed 42) and a single injection layer; whether the hallucination pattern (No-growth noise doubling HiConfErr) generalizes across seeds, layers, and architectures requires further investigation.

13 Conclusion

This paper establishes that non-normal operator geometry provides a principled framework for understanding the expressivity–stability tradeoff in deep networks, with direct implications for the hallucination problem. The key contributions are:

1. **Expressivity–Stability No-Go Theorem:** Transient amplification and excess noise sensitivity are inseparable consequences of non-normality. Forced normalization causes a 3% accuracy drop on CIFAR-10 ($p = 0.006$) and a 19% PPL degradation on WikiText-103 ($p = 0.0006$), while simultaneously producing the lowest confidence on errors ($p = 0.0021$).
2. **Curvature as the controlling quantity:** The curvature $R(T) = [H, K]$ strongly predicts excess noise ($r = 0.91$) and transient growth ($r = 0.59$). On a trained Transformer, curvature grows 3.2–4.6× during training and correlates with excess noise at $r = 0.77$ across layers.
3. **PRC as a non-normality probe:** Constructive PRC ($\beta \geq 0$) reduces LM perplexity by 1.38 ($p = 0.016$, secondary). Destructive PRC ($\beta \in [-1, 1]$) reveals layer-specific noise structure—4/6 layers learn $\beta < -0.1$ in every seed on language modeling vs. mostly $\beta \approx 0$ on CIFAR-10, confirming the task-dependent noise prediction.
4. **Curvature inflation:** Uniform suppression via negative β does not durably reduce non-normality. The model's weights compensate by developing higher curvature elsewhere, ultimately *increasing* confidence on errors (+0.013, $p = 0.0012$, Holm-significant across 3 seeds). This waterbed effect demonstrates that the optimal curvature is an emergent property of training dynamics.
5. **Post-hoc discrimination is insufficient; training-time discrimination works.** A targeted noise discriminator produces a null result when applied post-hoc to frozen weights (Group G: gate $\to 0$, ConfOnErr unchanged), revealing tight signal–noise entanglement. When co-trained with the base model (Group H), the same discriminator architecture reduces ConfOnErr by 20.1% ($p = 0.0001$, Holm-significant) and HiConfErr by 52.5% ($p = 0.0010$) with no PPL penalty. The gate again decays to zero, but the effect persists through shaped representations—the discriminator acts as a transient training signal. The ECE tradeoff (0.056 vs. 0.009) remains an open issue.
6. **Noise geometry determines the hallucination pattern.** Direct noise injection into a trained model reveals that the most dangerous noise is not the most non-normal but the most *spectrally persistent*: weakly non-normal noise without transient amplification doubles high-confidence errors (HiConfErr: 1.25% $\to$ 2.73%), while highly non-normal noise with transient growth reduces them by 86% (HiConfErr: 1.25% $\to$ 0.18%). Transient amplification serves a dual role: it amplifies noise (harmful) but also makes the perturbation detectable (protective). The most insidious noise regime is persistent but silent.

The central message is that hallucination in deep networks has a *geometric component* rooted in the non-normal structure of layer computations. Uniform methods (PRC, regularization) cannot selectively suppress amplified noise because the model compensates (curvature inflation). Post-hoc methods (discriminator gating) cannot disentangle signal from noise in frozen representations because the No-Go theorem guarantees they share the same amplifica-

tion pathways. Training-time co-discrimination offers a viable path: by shaping representations during learning, it achieves lower confidence on errors than even the forced-normal model—without sacrificing expressivity. The noise injection experiment adds a final, surprising insight: transient amplification is not purely harmful—it provides an implicit detection signal that helps the model recalibrate its confidence. The most dangerous noise regime is spectrally persistent but non-amplifying, producing the classic hallucination pattern of high-confidence errors.

Code and data availability

All training code, analysis scripts, and model checkpoints for reproducing the experiments in this paper are available at `https://github.com/zetacheng/nonnormal-geometry`[1]. The repository includes: (1) Phase 1–2 training scripts for Groups A–D, G, and H on both CIFAR-10 and WikiText-103; (2) Phase 3 analysis code producing all tables, figures, and statistical tests; (3) Phase 4 noise injection code with pre-constructed noise matrices; (4) pre-trained checkpoints for all groups and seeds. WikiText-103 is publicly available via HuggingFace; CIFAR-10 is available via `torchvision`.

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[1] Checkpoints available on request due to file size (∼120 MB each).

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